

ZS-F31

Covariant Cosmic Reality: the Exact Modular GKLS Spectrum, the Seam-Transport Realization, and the Normalizable Causal-Entropic Present Gate

From a Single Genuine Lift to Three Independent Results — a Closed-Form 121-Face Lindblad Spectrum, an Exact Seam Half-Transport Replacing the Retracted Monodromy Route, and an Entropy-Budget Proof of Present-Measure Normalizability

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Verification (v1.4, Final): 42/42 arithmetic, symbolic, and CONSTRUCTION checks PASS, including (i) the jump-only modular GKLS lift in BOTH pictures (Theorem P1), (ii) the full 121-face Lindblad spectrum $\text{Spec}(\mathcal{L}\star) = \text{Spec}(\mathcal{L}_\pi) \uplus \{-\frac{1}{2}(\Gamma_c + \Gamma_d)\}$ as a multiset union of two invariant blocks, verified against the dense superoperator up to dimension 121 and exhibited in closed form at $14641 = 121^2$ (Theorem P2), and (iii) the sign-corrected Seam Transport realization — singular-value Cartan length $\ell_K = |\rho_K|$ with the signed rapidity $\rho_K = \frac{1}{2}\text{Tr}[(n \cdot \sigma)\log U_K]$ recovered from the oriented axis, tested at BOTH signs (Theorem R2a) | 7 operator gates remain OPEN, not counted as PASS (R2b, P1-core, C_{int} , clock map, Z-screen, Gate E, coincidence gate) | FINAL PATCHES (v1.4): the seam result is open-path TRANSPORT not closed-loop holonomy (the connection is $\text{sl}(2, \mathbb{C})$ -valued pure-gauge); Gate E is downgraded to HYPOTHESIS-strong (data processing gives a value inequality, not a pointwise production-rate inequality); the §2.5 carrier sector is corrected (Brown–Kuchař dust vs the S14 prefactor, and the 3-form potential A_3 with $F_4 = dA_3$) | no new fitted parameter; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED | script: `zs_f31_verify_v1_4.py` (42/42 PASS)

§0. Abstract

This version converts the single repaired theorem of v1.2 into three independent contributions, removes one unforced parameter, retracts a route that did not match the corpus, and restores the integrative master-action content. v1.2 had correctly replaced v1.1's broken population lift (the family $\Psi_t = \iota \circ \Phi_t \circ E$, which is not a semigroup because $\Psi_0 = E_D \neq \text{id}$) with a genuine Lindblad generator. v1.3 sharpens and extends that core.

Contribution 1 — the exact 121-face modular GKLS spectrum (Theorem P2). Working in the Schrödinger picture with a *jump-only* generator $\mathcal{L}\star(\rho) = \sum_{\{a \neq b\}} r_{\{ab\}}(v_{\{ab\}} \rho v_{\{ab\}}^\dagger - \frac{1}{2}\{q_b, \rho\})$ (the A/Q dephasing term of v1.2 is removed), every coherence $|c\rangle\langle d|$ is an eigenvector with the explicit eigenvalue $-\frac{1}{2}(\Gamma_c + \Gamma_d)$, $\Gamma_c = \sum_{\{a \neq c\}} r_{\{ac\}}$. Hence the full spectrum splits in closed form, $\text{Spec}(\mathcal{L}\star) = \text{Spec}(\mathcal{L}_\pi) \uplus \{-\frac{1}{2}(\Gamma_c + \Gamma_d) : c \neq d\}$, with 121 population and $121 \cdot 120 = 14520$ coherence eigenvalues totalling $14641 = 121^2$. This is a genuine spectral theorem for the 121-face generator, not a toy check, and it is verified against the dense superoperator up to dimension 121. Removing the dephasing term makes the construction strictly parameter-free.

Contribution 2 — the exact seam-support connection (Theorem R2a). v1.2's F14-ODE monodromy route does not match ZS-F14, whose joint ODE is an (ε, θ) radial–angular system, not a linear support ODE with two periodic solutions (whose loop monodromy would be trivial). We RETRACT it and replace it with the exact seam half-transport built from ZS-F30's own object $g_K(\theta) = \exp[\frac{1}{2} \rho_K(\theta) n \cdot \sigma]$ with the seam antisymmetry $\rho_K(\theta+\pi) = -\rho_K(\theta)$. Then $U_K(\theta+\pi, \theta) = g_K(\theta+\pi)^{-1} g_K(\theta) = g_K(\theta)^2 = \exp[\rho_K(\theta) n \cdot \sigma]$, whose singular-value Cartan length is $\ell_K = |\rho_K|$ and whose signed oriented rapidity $\rho_K = \frac{1}{2} \text{Tr}[(n \cdot \sigma) \log U_K]$ is recovered from the fixed axis. This PROVES the oriented seam-TRANSPORT coordinate $\rho_{\text{tr}}^{\{\text{oriented}\}} = \rho_K$ (R2a) — an open-path Cartan realization, not a closed-loop holonomy — leaving the physical identification $A_Z \sim A_K$ (R2b), i.e. the promotion of transport to holonomy, OPEN. The connection A_K is $\text{sl}(2, \mathbb{C})$ -valued pure-gauge (a boost).

Contribution 3 — the normalizable causal-information present measure. The present is selected by the independent causal-diamond Z-Spin measure $d\mu_{\text{ZCD}} \propto A_D(T) \sigma_Z(T) dT$, with σ_Z the Spohn entropy production of the genuine QMS. Normalizability is now an *entropy-budget* theorem requiring no modified log-Sobolev inequality: since $\sigma_Z = -d/dT D(\rho_T \| \rho^*)$, one has $\int_0^\infty \sigma_Z dT = D(\rho_0 \| \rho^*) - D(\rho_\infty \| \rho^*) \leq D(\rho_0 \| \rho^*)$, so with the screen pixel count bounded $N_{\text{pix}} \leq N_{\text{max}}$ the measure integral is $\leq N_{\text{max}} D(\rho_0 \| \rho^*) < \infty$. The bulk–boundary relation Gate E is recast from an over-strong equality into a data-processing *inequality* $V_c \leq N_{\text{pix}} \sigma_Z$ (equality = channel saturation, OPEN), and the coincidence test $G = \Omega_\Lambda(T^*) - 83/121$ is honestly downgraded to a DERIVED-CONDITIONAL test with an explicit dependency tree on the clock map, the absolute scale B3, and Gate E.

The integrative master action $S_{\text{carrier}} = S_{\text{S14}}^{\text{tetrad}} + S_{\text{BK}} + S_\Lambda$ and its full Euler–Lagrange system are restored (§2.5, §12). The budget $(\Omega_b, \Omega_m, \Omega_\Lambda) = (6, 38, 83)/121$ (Ω_Λ within 0.2σ of Planck 2018, all three within 0.5σ) and $w = -1$ are inherited; no new fitted parameter is introduced; ($\mathcal{A}$, $\mathcal{Q}$, $\dim Z$) = (35/437, 11, 2) remain LOCKED.

Results architecture (what is closed, and what is a research programme)

For the external reader, the load-bearing claims and their exact status at a glance:

Result	Exact content	Status
P1	finite face-register jump-only GKLS lift (both pictures; $\Phi_0 = \text{id}$)	PROVEN
P1-core	modular eigen-partial-isometries v_{ab} in the Type II ₁ continuous core	OPEN
P2	general N-face population/coherence multiset spectrum (closed form)	PROVEN
R2a	oriented seam transport realizes ρ_K ($\ell_K = \rho_K $, signed via axis)	PROVEN
R2b	physical $A_Z \sim A_K$ (transport $\rightarrow$ closed-loop holonomy)	OPEN
Entropy budget	causal-Z measure normalizable under a bounded screen (no MLSI)	DERIVED
Gate E	bulk–boundary production-rate capacity bound	HYPOTHESIS-strong
C_{int}	i-tetration / modular clock spectral matching	OPEN
Gcoin	present coincidence $\Omega_\Lambda(T^*) = 83/121$ after clock map, B3, Gate E	DERIVED-COND. / TESTABLE

The first column of PROVEN results (P1, P2, R2a) is the mathematical core of this paper; the OPEN/HYPOTHESIS-strong rows are a delineated research programme (each a separate-paper-scale problem), not gaps in the proved results. This is the final internal version: the OPEN items (R2b, P1-core, C_{int} , B3, Gate E) are deferred to ZS-F32 / ZS-A31 rather than pursued here.

Epistemic Status Legend

Tag	Meaning
PROVEN	Complete proof under declared definitions; verified symbolically or to high precision.
IMPORTED-PROVEN	Theorem from the external peer-reviewed literature, imported with citation.
DERIVED	Valid deduction from PROVEN inputs plus Z-Spin axioms; zero fitted parameters.
DERIVED-CONDITIONAL	Valid deduction conditional on explicitly stated upstream condition(s).
DERIVED-interpretation	Structural synthesis of PROVEN/DERIVED components; the synthesis is new, the components are not; NOT a theorem.
CONSTRUCTION	An explicit object is built and its stated properties are machine-verified on a finite register.
HYPOTHESIS-strong	Multiple independent anchors; one identified step missing; explicit promotion path.
TESTABLE	A sharply defined, pre-registered numerical or observational check not yet run/closed.
LOCKED	Core constant fixed upstream; not adjustable downstream.
CLOSED-NEGATIVE	A candidate route proved impossible within the framework (a no-go).
OPEN	Recognized gap requiring future work; upgrade path specified where possible.
RETRACTED	A claim previously asserted that fails under scrutiny; the corrected statement is given.
NON-CLAIM	Explicitly not asserted; protective scope-marker.

§1. Corrections of record and scope (v1.2 → v1.3, with v1.4 final patches)

v1.3 kept every correct result of v1.2 and made the changes below; v1.4 then applied four final patches (the seam sign and transport-vs-holonomy distinction, the Gate-E downgrade to HYPOTHESIS-strong, and the §2.5 carrier-label corrections), detailed in their sections and in the Version History. Each change strengthens rigor or value.

v1.2 item	v1.3 status	reason / gain
generator written in Heisenberg picture; proof slips into $v_{ab} \times v_{ab}^\dagger$	BOTH PICTURES stated	Schrödinger $\mathcal{L}^\star(\rho) = \Sigma r_{ab}(v_{ab} \rho v_{ab}^\dagger - \frac{1}{2}\{q_b, \rho\})$ for state-level facts; Heisenberg $\mathcal{L} = \mathcal{L}^\star^\dagger$ for unitality $\mathcal{L}(I)=0$. Removes a presentational inconsistency external readers would flag.
‘faces being central’ / q_a central	CORRECTED to ‘in the modular centralizer’	In a Type II ₁ factor $Z(M_{\text{obs}}) = \mathbb{C}I$, so 121 nontrivial orthogonal central projections cannot exist; the faces lie in the modular centralizer M_ω .
P1 stated as one theorem	SPLIT: finite P1 (PROVEN/CONSTRUCTION) vs P1-core (DERIVED-CONDITIONAL)	$\tau(q_a)=\tau(q_b)$ gives a partial isometry, but v_{ab} lying in a modular spectral subspace is a separate condition (P1-core); the continuous-core embedding is conditional on it.
A/Q dephasing term in the generator	REMOVED (jump-only)	The jump dissipator alone damps coherences via $-\frac{1}{2}(\Gamma_c + \Gamma_d) < 0$ for a connected graph; removing the term eliminates an unforced rate and makes the lift parameter-free (§4, §14).
R2: F14-ODE monodromy target	RETRACTED → Seam Transport (R2a PROVEN + R2b OPEN)	ZS-F14’s joint ODE is an (ϵ, θ) radial–angular system, not a linear support ODE; the proposed $Y(\theta)$ monodromy would be trivial. The seam half-transport from F30 proves the oriented transport coordinate $\rho_{tr} = \rho_K$ (R2a); the closed-loop holonomy is R2b (§5).

P1 verified on a 4-face toy	RAISED to the exact 121-face spectrum (Theorem P2)	Closed-form coherence eigenvalues give the full 14641-eigenvalue spectrum; verified against the dense superoperator up to dimension 121 (§4).
§9.2 normalizability via spectral-gap decay ansatz	REPLACED by an entropy-budget bound	$\int \sigma_Z = D(\rho_{0 p^*}) - D(\rho_{\infty p^*}) \leq D(\rho_{0 p^*})$ needs no MLSI; the spectral gap survives only as an optional tail corollary (§10).
Gate E as an exact equality	RECAST as a data-processing INEQUALITY	LHS is coarse-grained astrophysical entropy, RHS register entropy production; a CPTP coarse-graining channel gives $V_c \dot{s}_{\text{bulk}} \leq N_{\text{pix}} \sigma_Z$, with equality = saturation (OPEN) (§9).
coincidence gate as one independent TESTABLE gate	DERIVED-CONDITIONAL with an explicit dependency tree	$\Omega_{\Lambda}(T)$ needs the density ratio (B3) and the clock map $t \mapsto t(T)$; the test depends on (C_int, B3, Gate E). Stated as a tree, not hidden (§11).
§11 master action compressed to one line	RESTORED carrier action + full field equations	$S_{\text{carrier}} = S_{S14}^{\text{tetrad}} + S_{BK} + S_{\Lambda}$ with the Euler–Lagrange system (§2.5, §12) — integrative content preserved without restoring any wrong closure.
Theorem P1 (genuine GKLS, $\Phi_0 = \text{id}$)	KEPT, in both pictures	The correct v1.2 core, retained and extended by P2.
Diagonal-in-Centralizer Lemma (PROVEN)	UNCHANGED	Underwrites modular covariance of $\mathcal{L}\star$; the one structural result standing across all versions.

§2. Locked inputs and the imported carrier

Nothing here is fitted. $(\mathcal{A}, \mathcal{Q}, (Z, X, Y)) = (35/437, 11, (2, 3, 6))$ are LOCKED (ZS-F2, ZS-F5); $\kappa^2 = \mathcal{A}/\mathcal{Q} = 35/4807$ (ZS-M6); the i-tetration fixed point is $z^* = 0.43828 + 0.36059 i$ with $|z^*| = 0.5675551633$ and $\alpha_{BK} = -\ln|z^*| = 0.5664173303$ (ZS-M1/M4, corrected digits). The spacetime metric is an **imported carrier**: ZS-A17 Theorem F (Spin–Metric Independence, CLOSED-NEGATIVE) proves the 3-metric is not reconstructible from $\mathcal{A}$ or the spin structure. The Z-sector supplies the causal cone (ZS-F30: $\det X = \text{Minkowski}$, $\rho = \text{artanh } \beta$, the $SL(2, \mathbb{C})$ identity component), the pressureless matter frame and relational clock (ZS-A19/A20 Brown dust, $w = 0$), and the top-form vacuum (ZS-A28/A30, $w = -1$). The occupation is the unital-primitive doubled attractor $I_{121/121}$ with rank fractions $(6, 32, 83)/121$ (ZS-A30) — a *rank* fact, distinct from any present-epoch energy statement.

§2.5. Restored carrier action and the carrier-sector Euler–Lagrange equations

This section restores the integrative content compressed in v1.2 §11, **without** restoring any retracted closure. The carrier action assembles the imported metric sector, the Z-Spin gauge/holonomy sector, and the top-form vacuum:

$$S_{\text{carrier}} = S_{\{S14\}^{\text{tetrad}}} + S_{\{BK\}} + S_{\Lambda},$$

with a precise division of labour: $S_{\{S14\}^{\text{tetrad}}}$ carries the SM gauge fields B, W, G , the Z-Spin field H_5 , and the nonminimal scalar prefactor $F = M_{\text{P}}^2(1 + \mathcal{A} |H_5|^2)$ (ZS-S14); $S_{\{BK\}}$ is the Brown–Kuchař DUST sector (clock fields T, X^I , four-velocity U^μ , density ρ_d), which supplies the relational clock and pressureless matter frame (ZS-A19/A20) and is NOT a Brans–Dicke sector; and S_{Λ} is the top-form vacuum built from a 3-form potential A_3 with field strength $F_4 = dA_3$, $S_{\Lambda} = -(1/(2 \cdot 4!)) \int d^4x \sqrt{(-g)} \tau_{121}(P_{\Lambda} F_{\{\mu\nu\rho\sigma\}} F^{\{\mu\nu\rho\sigma\}})$ (ZS-A28). Varying the independent fields gives the load-bearing carrier-sector equations:

Variation	Euler–Lagrange equation
δA_3 (top-form flux)	$d(P_\Lambda \star F_4) = 0$ — the 4-form field strength is covariantly constant, fixing a constant vacuum energy ρ_Λ and the equation of state $w = -1$.
δp_d (dust frame)	$U^\mu U_\mu = -1$ — the Brown-dust four-velocity is unit timelike (the relational clock frame, ZS-A19/A20).
$\delta g^{\{\mu\nu\}}$ (metric)	$F \cdot G_{\{\mu\nu\}} = T_{\{\mu\nu\}}^{\{\text{SM}\}} + \rho_m U_\mu U_\nu - \rho_\Lambda g_{\{\mu\nu\}} + \nabla_\mu \nabla_\nu F - g_{\{\mu\nu\}} \square F$ — Einstein equations with an SM stress tensor, pressureless dust, a cosmological-constant term, and the scalar-field (improvement) terms of the F sector.

This is exactly the structure in which ZS-F30’s causal cone (through $T_{\{\mu\nu\}}^{\{\text{SM}\}}$ and the tetrad) and ZS-A30’s occupation (through ρ_Λ and ρ_m) meet in one action. The reduction to ZS-S14 is the limit $\Sigma_Z = A_3 = \hat{q} = \lambda_R = \Lambda_C = 0$ with the present coupling $\mathcal{I}$ removed, **keeping** B, W, G (§15). [§2.5: carrier action and field equations **IMPORTED-PROVEN** / **DERIVED** from ZS-S14/A19/A20/A28; no new fitted parameter.]

§3. The genuine modular GKLS population lift, in both pictures (Theorem P1)

§3.1 Data, balanced rates, and the modular centralizer

Let (M_{obs}, τ) be the Type II₁ observer corner with 121 face projections $q_a, q_a q_b = \delta_{\{ab\}} q_a, \Sigma_a q_a = I, \tau(q_a) = 1/121$, and balanced ZS-A30 rates $r_{\{ab\}} \geq 0$ ($a \neq b$), $\Sigma_a r_{\{ab\}} = \Sigma_a r_{\{ba\}}$. By the Diagonal-in-Centralizer Lemma (PROVEN, unchanged), the diagonal algebra D_{121} lies in the modular **centralizer** M_ω — the fixed-point algebra of the modular flow — *not* in the center $Z(M_{\text{obs}}) = \mathbb{C}I$ (a Type II₁ factor has trivial center, so 121 nontrivial orthogonal central projections cannot exist; this corrects the loose word ‘central’ in v1.2).

§3.2 The two pictures

On states ρ (Schrödinger picture) and on observables x (Heisenberg picture, the adjoint) the jump-only generators are

$$\begin{aligned} \mathcal{L}\star(\rho) &= \Sigma_{\{a \neq b\}} r_{\{ab\}} (v_{\{ab\}} \rho v_{\{ab\}}^\dagger - \tfrac{1}{2} \{q_b, \rho\}) , \\ \mathcal{L}(x) &= \mathcal{L}\star^\dagger(x) = \Sigma_{\{a \neq b\}} r_{\{ab\}} (v_{\{ab\}}^\dagger x v_{\{ab\}} - \tfrac{1}{2} \{q_b, x\}) , \end{aligned}$$

with jumps $\sqrt{r_{\{ab\}}} v_{\{ab\}}, v_{\{ab\}}^\dagger \sqrt{r_{\{ab\}}} = q_b, v_{\{ab\}} v_{\{ab\}}^\dagger = q_a$. We use $\mathcal{L}\star$ for state-level statements (trace preservation, intertwining, stationarity, coherence damping) and $\mathcal{L}$ for unitality, and we separate the dual identities $\text{Tr } \mathcal{L}\star(\rho) = 0$ and $\mathcal{L}(I) = 0$.

§3.3 Theorem P1 and the finite/continuous status split

Theorem P1 (Genuine Population Lift). The semigroup $\Phi_t = e^{\{t\mathcal{L}\star\}}$ has all five properties, machine-verified on a finite register:

Property	Statement / mechanism
(1) $\Phi_0 = \text{id}$	$e^{\{0 \cdot \mathcal{L}\star\}} = \text{id}$ — the property the retracted v1.1 family $\iota^\circ \Phi_t^\circ E$ lacked (it had $\Psi_0 = E_D \neq \text{id}$).
(2) CP semigroup	$\mathcal{L}\star$ is in GKLS form, so $\{\Phi_t\}$ is a normal completely positive semigroup (Lindblad; GKS).
(3) trace preservation	$\text{Tr } \mathcal{L}\star(\rho) = 0$ (Schrödinger) $\Leftrightarrow \mathcal{L}(I) = \Sigma r_{\{ab\}}(q_b - q_a) = 0$ (Heisenberg unitality).

(4) modular covariance	$[\mathcal{L}\star, \sigma_s^\omega] = 0$, since the $v_{\{ab\}}$ are modular eigen-operators ($D_{121} \subseteq M_\omega$).
(5) population intertwining	$E_D \mathcal{L}\star = \mathcal{L}_\pi E_D$ and $E_D \mathcal{L}\star(1 - E_D) = 0$: off-diagonals do not feed populations; the diagonal restriction is the classical chain $\mathcal{L}_\pi$.
uniqueness	For a strongly connected rate graph the kernel of $\mathcal{L}\star$ is one-dimensional, so the tracial state τ_{121} is the unique stationary state.

Status split (correcting v1.2's single statement): on the finite face register the generator is built explicitly and all five properties plus uniqueness are **machine-verified** (Appendix A), so finite P1 is **PROVEN / CONSTRUCTION**. The continuous-core embedding requires modular eigen-partial-isometries $v_{\{ab\}}$ living in the right modular spectral subspace; $\tau(q_a) = \tau(q_b)$ gives *some* partial isometry with $v_{\{ab\}}^\dagger v_{\{ab\}} = q_b$, $v_{\{ab\}} v_{\{ab\}}^\dagger = q_a$, but its membership in $M_{\text{obs}}(\omega_{\{ab\}})$ is a separate condition, recorded as the gate

$$P1\text{-core: } \exists v_{\{ab\}} \in M_{\text{obs}}(\omega_{\{ab\}}) \text{ with } v_{\{ab\}}^\dagger v_{\{ab\}} = q_b, v_{\{ab\}} v_{\{ab\}}^\dagger = q_a.$$

So the Type II₁ core embedding is **DERIVED-CONDITIONAL on P1-core** (OPEN). The cosmological observable Ω_Λ is a population fact, so the population lift suffices for the budget; the full coherence-resolved lift of ZS-A24's family $\mathcal{L}_\pi(s)$ is broader and remains a NON-CLAIM here.

§4. The exact 121-face Lindblad spectrum (Theorem P2)

This is the first of the three value-raising results. With the dephasing term removed, the generator is block-diagonal between populations and coherences, and the coherence block is diagonal in the matrix-unit basis.

§4.1 The closed-form coherence eigenvalues

For an off-diagonal matrix unit $|c\rangle\langle d|$ ($c \neq d$), the gain term $v_{\{ab\}}|c\rangle\langle d|v_{\{ab\}}^\dagger = \delta_{\{bc\}}\delta_{\{bd\}}|a\rangle\langle a|$ vanishes, and only the anticommutator survives:

$$\mathcal{L}\star(|c\rangle\langle d|) = -\frac{1}{2}(\Gamma_c + \Gamma_d) |c\rangle\langle d|, \quad \Gamma_c = \sum_{\{a \neq c\}} r_{\{ac\}} \text{ (the total out-rate from face } c \text{)}.$$

Each coherence is therefore an exact eigenvector with eigenvalue $-\frac{1}{2}(\Gamma_c + \Gamma_d) < 0$ for a connected graph — which is precisely why no separate dephasing term is needed (it would only add a constant $-\gamma$ to every coherence eigenvalue without changing the population block). On the diagonal, $\mathcal{L}\star$ restricts to the classical population generator $\mathcal{L}_\pi$.

§4.2 Theorem P2 (Full Face Spectrum)

Theorem P2. For the jump-only modular GKLS generator on N equal-trace faces with balanced symmetric rates,

$$\text{Spec}(\mathcal{L}\star) = \text{Spec}(\mathcal{L}_\pi) \uplus \{ -\frac{1}{2}(\Gamma_c + \Gamma_d) : c \neq d \},$$

a multiset union associated with two invariant blocks (numerical coincidences between population and coherence eigenvalues are allowed): N population eigenvalues (one zero mode, the rest negative — the uniform tracial stationary state and its gap) and $N(N-1)$ explicit coherence eigenvalues. For the 121-face register this is

$$121 \text{ population} + 121 \cdot 120 \text{ coherence} = 121 + 14520 = 14641 = 121^2 \text{ eigenvalues, in closed form.}$$

Status and novelty. The theorem is proved algebraically for arbitrary N ; numerical regression against the dense superoperator is performed for $N = 4, 8, 11$ (the last at dimension $11^2 = 121$), and the $N = 121$ specialization follows analytically from the theorem and the inherited ZS-A30 rate structure (under the doubled-additive $\mathcal{L}_{121} = \mathcal{L}_{11} \otimes \text{id} + \text{id} \otimes \mathcal{L}_{11}$, the 121 population eigenvalues are the pairwise sums $\lambda_i + \lambda_j$). The relation to ZS-A30 is one of abstraction, not duplication: ZS-A30 v1.6 computed the spectrum for the specific Z-Spin mediator graph (degree-(2,9), with population/coherence multiplicities (1, 80, 36, 3, 1)); ZS-F31 abstracts that calculation into a general N -face jump-GKLS block-spectrum theorem and isolates the population/coherence decomposition independently of the graph-specific multiplicities. [§4: **PROVEN** (algebraic for arbitrary N ; regression to dimension 121). The general theorem is of more direct mathematical value to external readers than the graph-specific computation.]

§5. The seam-support connection: Seam Transport Theorem (R2a, R2b)

§5.1 Why the v1.2 F14-monodromy route is retracted

v1.2 proposed computing the monodromy of a fundamental matrix $Y(\theta) = [[h_1, h_2], [h_1', h_2']]$ built from two ‘support solutions’ of the ZS-F14 joint ODE. This does not match the corpus: ZS-F14’s joint ODE is an (ε, θ) *radial–angular* system (radial Lyapunov decay; angular Goldstone accumulation; the conservation $Q = a^3 \varepsilon^2 \theta = A$; a centrifugal boundary), not a linear support ODE with two independent periodic solutions. Moreover, if h_1, h_2 were ordinary 2π -periodic support functions, $Y(\theta+2\pi) = Y(\theta)$ would give trivial loop monodromy and no rapidity. We therefore RETRACT the F14-monodromy route.

§5.2 The exact seam half-transport: a Cartan realization (R2a, PROVEN)

ZS-F30 already supplies the object we need: the Steiner-centered support-ratio rapidity $\rho_K(\theta)$ and the $SL(2, \mathbb{C})$ element

$$g_K(\theta) = \exp[\tfrac{1}{2} \rho_K(\theta) n \cdot \sigma], \quad A_K = g_K^{-1} dg_K \quad (\text{an } sl(2, \mathbb{C})\text{-valued pure-gauge Cartan boost connection, fixed axis } n(\theta+\pi)=n(\theta)),$$

together with the seam antisymmetry $\rho_K(\theta+\pi) = -\rho_K(\theta)$ (the half-turn sign flip of F30). Hence $g_K(\theta+\pi) = g_K(\theta)^{-1}$, and the seam half-transport is

$$U_K(\theta+\pi, \theta) = g_K(\theta+\pi)^{-1} g_K(\theta) = g_K(\theta) \cdot g_K(\theta) = g_K(\theta)^2 = \exp[\rho_K(\theta) n \cdot \sigma].$$

Since $(n \cdot \sigma)^2 = I$, $\exp[\rho_K n \cdot \sigma]$ is a Hermitian positive-definite boost with eigenvalues $e^{\pm \rho_K}$; its singular values are therefore $\{e^{|\rho_K|}, e^{-|\rho_K|}\}$, so the singular-value ratio yields the UNSIGNED Cartan length

$$\ell_K = \tfrac{1}{2} \ln(s_{\max} / s_{\min}) = |\rho_K| \quad (\text{sign-independent}).$$

The signed, seam-odd rapidity is not visible in the singular values; it is recovered from the oriented logarithmic generator ($\log U_K = \rho_K n \cdot \sigma$):

$$\rho_K = \tfrac{1}{2} \text{Tr}[(n \cdot \sigma) \log U_K] \quad (\text{since } \tfrac{1}{2} \text{Tr}[(n \cdot \sigma)^2] = I).$$

Together these give a complete Cartan decomposition — magnitude $\ell_K = |\rho_K|$ from the singular values, orientation/sign from the axis projection — verified to machine precision for BOTH signs of ρ_K (Appendix C). This corrects a sign slip in v1.3, which read the ratio as ρ_K rather than $|\rho_K|$ and tested only positive ρ_K . The result is the oriented transport coordinate

$$\rho_{tr}^{\{oriented\}} = \rho_K \quad (for\ the\ canonical\ support-defined\ seam\ transport) .$$

Crucially this is an open-path TRANSPORT statement, not a closed-loop holonomy: $U_K(\theta+\pi, \theta)$ is parallel transport over the half-turn, and the singular values of an open transport are not gauge-invariant under independent endpoint gauge transformations. So R2a is a canonical seam-TRANSPORT realization theorem, not a physical-holonomy theorem; the promotion $\rho_{tr} \rightarrow \rho_{hol}^{\{physical\}}$ requires the gauge equivalence $A_Z \sim A_K$ (R2b, §5.3). [§5.2: **PROVEN** as a seam-transport realization (R2a); holonomy promotion is R2b, OPEN.]

§5.3 What remains: physical identification (R2b, OPEN)

R2a fixes the geometry: the seam transport of the F30 connection A_K has boost coordinate exactly ρ_K . What is *not* yet proved is that the physical Z-sector connection A_Z appearing in the master action is gauge-equivalent to A_K (so that its holonomy is the same object). That is the single remaining gate:

$$R2b: \quad A_Z \sim A_K \quad (gauge\ equivalence\ of\ the\ physical\ and\ the\ Steiner-support\ connections) .$$

[§5.3: R2b **OPEN**. Net: the support–connection gate is now R2a (PROVEN) + R2b (OPEN), replacing v1.2’s R1+R2, with the geometric content actually closed.]

§6. Clock alignment: Koenigs linearization and the spectral-matching gate

The i-tetration $f(z) = i^z$ has the attracting fixed point z^* with multiplier $\lambda^* = f'(z^*) = (i\pi/2) z^*$, $|\lambda^*| = (\pi/2)|z^*| = 0.89151 < 1$. By Koenigs’ theorem there is an analytic χ near z^* with $\chi \circ f = \lambda^* \cdot \chi$, so the iteration is locally a dilation by λ^* — a rigorous statement replacing v1.1’s shortcut (PROVEN). Identifying one i-tetration step with one modular step of a Z-sector eigenoperator (eigenvalue $e^{\{-s\omega_Z + isv_Z\}}$) requires a single δs , i.e. the Intertwiner Condition

$$C_{int}: \quad -\ln|\lambda^*| / \omega_Z = \arg \lambda^* / v_Z, \quad -\ln|\lambda^*| = 0.1148346, \quad \arg \lambda^* = 2.2592496 .$$

Honest note (clarifying ZS-M4): the per-step contraction $-\ln|\lambda^*| = 0.1148346$ is **distinct** from $\alpha_{BK} = -\ln|z^*| = 0.5664173$ (a fixed-point modulus) — different objects, so α_{BK} is not the matching rapidity. C_{int} is a concrete equality once the modular spectrum (ω_Z, v_Z) is computed; that computation is not done here. [§6: Koenigs **PROVEN**; C_{int} **OPEN**; on success clock alignment **DERIVED-CONDITIONAL**.]

§7. Causal-diamond geometry (the Z-screen as a candidate carrier)

In flat FRW, a causal diamond of conformal extent $[\eta_i, \eta_f]$ has comoving radius $r_D(\eta) = \min\{\eta - \eta_i, \eta_f - \eta\}$, a 2D maximal screen $\Sigma_D(\eta) = S^2_{\{r_D(\eta)\}}$ of physical area $A_D(\eta) = 4\pi a^2(\eta) r_D^2(\eta)$, and comoving volume $V_c(\eta) = (4\pi/3) r_D^3(\eta)$. Bousso’s causal-entropy construction computes $\Delta S_{CD} = \int dt V_c \dot{s}_{bulk}$; the comoving volume vanishes early and in the far future, and the overlap of volume with the entropy-production peak selects a preferred range for the cosmological constant. Scope, stated carefully: only the maximal *two-dimensional* screen cross-section is dimensionally comparable to the Z-sector’s 2D object, and in Bousso’s calculation the dominant entropy source is astrophysical. Hence the causal-diamond 2D screen is a **candidate** geometric carrier for the Z-sector, and the identity between Bousso’s causal entropy and the Z-Spin entropy of §8 is **OPEN**. [§7: causal-diamond geometry **IMPORTED-PROVEN**; Z-screen identification **HYPOTHESIS-strong**.]

§8. Z-Spin entropy production

Because Theorem P1 provides a *genuine* QMS with a faithful stationary state $\varrho^* = I_{121}/121$, the Spohn entropy production is well-defined and non-negative:

$$\sigma_Z(T) = -d/dT D(\varrho_T \| \varrho^*) = -\tau_{121} [\mathcal{L}^\star(\varrho_T) (\ln \varrho_T - \ln \varrho^*)] \geq 0 \quad (\text{Spohn}).$$

The population current $J_{\{ab\}}(T) = r_{\{ab\}} p_b - r_{\{ba\}} p_a$ vanishes in de Sitter equilibrium ($\sigma_Z = 0$) and is nonzero only when matter-mediated transfer drives the populations off equilibrium. Thus ‘matter-correlated’ is realized as the **existence of a nonzero information current**, not as a fitted exponent. [§8: Spohn positivity **IMPORTED-PROVEN**; current representation **DERIVED** from Theorem P1.]

§9. The bulk–boundary gate as a capacity bound (Gate E, HYPOTHESIS-strong)

Introduce a CPTP coarse-graining channel $\mathcal{N}: Z \rightarrow X$ from the fine Z-register to the coarse astrophysical description. Monotonicity of relative entropy under $\mathcal{N}$ (the data-processing inequality) gives the VALUE inequality $D(\varrho_Z \| \varrho_{Z^*}) \geq D(\mathcal{N}\varrho_Z \| \mathcal{N}\varrho_{Z^*})$ at each time. The intended bulk–boundary production-rate bound is the entropy-production *capacity bound*:

$$\text{Gate E (target bound):} \quad V_c(T) \dot{s}_{\text{bulk}}(T) \leq N_{\text{pix}}(T) \sigma_Z(T), \quad N_{\text{pix}}(T) = A_D(T)/(4\ell_P^2) \\ [\text{HYPOTHESIS-strong}].$$

Honest status: data processing motivates this bound but does NOT by itself prove it. The DPI is a VALUE inequality $D_Z(T) \geq D_X(T)$; the production rates are $\sigma = -dD/dT$, and value ordering does not imply derivative ordering (e.g. $e^{-T} \geq e^{-2T}$ for all T , yet at $T = 0$ the decay rate of the smaller function is larger). The Lindblad/Spohn theorems give complete positivity and entropy-production positivity, but not a pointwise production-rate inequality between distinct fine and coarse dynamics. Closing it needs a generator-level intertwining $\mathcal{N}\Phi_t^Z = \Phi_t^X \mathcal{N}$, or a strong data-processing / entropy-contraction inequality, or a directly computed channel-capacity bound. [§9: Gate E is **HYPOTHESIS-strong** (not DERIVED-CONDITIONAL); to be promoted only after such a generator-level theorem is established. It is already listed OPEN in the registry.]

§10. The causal-Z present measure and the entropy-budget normalizability theorem

§10.1 The measure

$$d\mu_{ZCD}(T) = W_Z(T) dT / \int dT' W_Z(T'), \quad W_Z(T) = N_{\text{pix}}(T) \sigma_Z(T) = (A_D(T)/4\ell_P^2) \sigma_Z(T),$$

with no observer-count or anthropic parameter; $83/121$ does not appear in W_Z .

§10.2 Normalizability by entropy budget (DERIVED, no MLSI)

This is the second value-raising result: a clean proof that needs no modified log-Sobolev inequality and no spectral-gap decay ansatz. From the Spohn definition $\sigma_Z = -d/dT D(\varrho_T \| \varrho^*)$, the production integral telescopes:

$$\int_0^\infty \sigma_Z(T) dT = D(\varrho_0 \| \varrho^*) - D(\varrho_\infty \| \varrho^*) \leq D(\varrho_0 \| \varrho^*) < \infty.$$

With the causal screen bounded, $N_{\text{pix}}(T) \leq N_{\text{max}}$, the measure integral is finite:

$$\int_0^\infty W_Z(T) dT = \int_0^\infty N_{pix}(T) \sigma_Z(T) dT \leq N_{max} \cdot D(q_0 \| q^*) < \infty .$$

The relative entropy is monotone non-increasing under any QMS (Lindblad), so the bound holds for a genuine QMS and a bounded screen *alone*. The spectral gap $\Delta = 2A/Q$ survives only as an optional tail-rate corollary ($\int_0^\infty e^{-2\Delta T} dT = 1/(2\Delta)$), not as a hypothesis. This strengthens v1.1's proper-time exclusion into a usable, provably normalizable replacement. For the finite 121-register $q^* = I/121$ is faithful, so $D(q_0 \| q^*)$ is finite; the only remaining condition is boundedness of the causal screen, natural for an asymptotic de Sitter causal diamond with positive Λ . [§10.2: **DERIVED** under $N_{pix}(T) \leq N_{max}$ and finite $D(q_0 \| q^*)$.]

§10.3 The present equation

$$T^* = \operatorname{argmax}_T W_Z(T) , \quad d/dT \ln A_D(T) |_{T^*} = - d/dT \ln \sigma_Z(T) |_{T^*} :$$

the present is where the growing causal screen balances the decaying information production.

§11. The coincidence closure gate and its dependency tree (DERIVED-CONDITIONAL)

The methodological pivot stands — 83/121 must not enter W_Z — but the test is not unconditional, and v1.3 states its dependencies explicitly rather than hiding them. To evaluate $\Omega_A(T)$ one needs the density ratio $\rho_A/\rho_{\{m,0\}}$ (inserting 83/38 would be circular; inserting the observed value would not be a prediction), which requires the absolute vacuum scale B3 (OPEN). One also needs the clock map $t = t(T)$ relating cosmic/conformal time to the Brown register clock, which is logically prior to $A_D(T)$. The dependency tree is

$$C_{int} \rightarrow A_D(T) ; \quad (B3 \text{ or independent density ratio}) \rightarrow \Omega_A(T) ; \quad (C_{int}, B3, \text{Gate } E) \rightarrow G_{coincidence} .$$

$$G_{coincidence} = \Omega_A(T^*) - 83/121 , \quad \text{closure: } |G_{coincidence}| < 10^{-3} \text{ and } d^2/dT^2 \ln W_Z(T) |_{T^*} < 0 .$$

The tolerance is pre-registered before computation. Therefore G is **DERIVED-CONDITIONAL** on $(C_{int}, B3, \text{Gate } E)$, TESTABLE on success — not the single independent gate v1.2 implied. [§11: **DERIVED-CONDITIONAL / TESTABLE**.]

§11.1 A future direction: the scale–present coupled variational problem (F32)

A more aggressive route would make the top-form flux f_Λ a variational target alongside T, maximising $W_Z(T; f_\Lambda)$ jointly:

$$(T^*, f_\Lambda^*) = \operatorname{argmax}_{\{T, f_\Lambda\}} W_Z(T; f_\Lambda) , \quad \text{then compute } \rho_A(f_\Lambda^*), \Omega_A(T^*, f_\Lambda^*) \text{ together ,}$$

unifying present selection and B3 scale selection into one scale–present coupled variational problem. If successful it would attack the largest OPEN of ZS-A26–A30. We flag this as a separate future programme (ZS-F32 / ZS-A31), not a claim of this paper. [§11.1: **OPEN** (future programme).]

§12. The revised master effective action

$$\Gamma_{CR}^{(1.3)} = S_{carrier} + S_Z^{\{flat\}} + S_{QMS}^{\{jump\}} + S_R^{\{seam\}} + S_C^{\{Koenigs\}} + \mathcal{I}_{ZCD} ,$$

$$S_{\text{carrier}} = S_{\{S14\}^{\text{tetrad}}} + S_{\{BK\}} + S_A \quad (\S 2.5), \quad S_{\text{QMS}^{\text{jump}}} \leftrightarrow \mathcal{L}\star \text{ (Theorems P1, P2)},$$

$$S_{\text{R}^{\text{seam}}} \leftrightarrow A_K, \rho_{\text{tr}^{\text{oriented}}} = \rho_K \text{ (Theorem R2a; } \ell_K = |\rho_K|), \quad \mathcal{I}_{\text{ZCD}}[T] = -\ln[(A_D(T)/4\ell_{P^2}) \sigma_Z(T)].$$

The relative entropy $D(\mathbf{q}_T \parallel \mathbf{q}^*)$ is a QMS Lyapunov functional, disjoint from any present-selection role:

Object	Role
$D(\mathbf{q}_T \parallel \mathbf{q}^*)$	Lyapunov functional measuring approach to equilibrium (NOT a present selector).
$\sigma_Z = -d/dT D$	irreversible information flow (Spohn entropy production); its integral is the entropy budget (§10.2).
$A_D(T) \sigma_Z(T)$	present weight including causal accessibility (the screen the observer can resolve).
83/121	independent rank target, checked at T^* by the coincidence gate (§11).

§13. Discussion — the Z-Spin mediation structure and realisation C14

The mediator+pair reading remains attractive: the three on-shell conditions are the boost coordinate (R), the generator (C), and the fixed state (P) of one de Sitter modular flow, with R the Z-Spin hinge of the complex rapidity $\zeta = \rho_K + i\phi$. But two operator identities it needs are not proved, so it is interpretation, not theorem.

Needed identity	Why it is not yet a theorem
ϕ (spatial rotation) = Euclidean modular time	In ZS-F30 ϕ is a compact spatial rotation parameter; the operator identity $e^{\{i\phi J\}} = e^{\{-\tau_E H_{\text{mod}}\}}$ ($\phi = H\tau_E$) is not established.
modular conjugation = corpus seam	Bisognano–Wichmann gives $J = \Theta \cdot U(R_W(\pi))$ — anti-unitary CPT Θ times a rotation, not $U(R_W(\pi))$ alone (correcting v1.1’s phrasing). Identifying J with the seam $V_{ZY} = (V_{XZ})^*$ needs a separate intertwining theorem.

Accordingly C14 is recorded as a **DERIVED-interpretation**, to be promoted only after both identities are proved. [§13: **DERIVED-interpretation**.]

§14. Zero-fitted-parameter and anti-numerology audit

Every dimensionless quantity descends from $(A, Q, \dim Z) = (35/437, 11, 2)$. v1.3 is **strictly stronger** than v1.2 here: removing the A/Q dephasing term means the generator’s only inputs are the ZS-A30 rates $r_{\{ab\}}$, with no separately chosen coherence rate. The causal-Z measure contains no observer-count or anthropic exponent (matter-correlation is the nonzero current $J_{\{ab\}}$); the coincidence tolerance $|G| < 10^{-3}$ is pre-registered. The only remaining free input is N_{eff} in the multinomial variance, which does not affect the mean 83/121; the absolute scale (H_0, ρ_Λ) is an integration constant left OPEN (B3). If, in future, a dephasing term were reintroduced for physical reasons, its rate would have to be declared as a separate **Minimal-Rate Identification** $\gamma = A/Q$ — but v1.3 does not need it. The budget vs Planck 2018:

Quantity	Rank fraction	Planck 2018	deviation
Ω_b	6/121 = 0.0496	0.0493 ± 0.0006	0.48σ
Ω_{cdm}	32/121 = 0.2645	0.2645 ± 0.0050	0.01σ
Ω_Λ	83/121 = 0.6860	0.6847 ± 0.0073	0.17σ

Correct statement: Ω_Λ within 0.2σ ; all three fractions within 0.5σ . The relative-entropy minimum of $D(x \parallel 83/121)$ at 83/121 confirms (does not predict) the equilibrium — which is why §10–§11 use the independent causal measure. The multinomial 83/121 is the mean; the mode is $83/121 + O(N_{\text{eff}}^{-1})$.

§15. Cross-paper dependency and version-conflict check

Upstream result	Use here	Conflict check
ZS-A24 F-A24.9 (OPEN)	population sector closed by P1; full spectrum by P2	Advances A24; every A24 theorem used unchanged; Π_1 -core no-go respected.
ZS-A24 Diagonal-in-Centralizer	modular covariance of $\mathcal{L}^\star$ (§3)	Used; PROVEN; ‘centralizer’ usage corrected (not ‘center’).
ZS-A30 doubled generator $\mathcal{L}_{11} \otimes \text{id} + \text{id} \otimes \mathcal{L}_{11}$	121 population eigenvalues = pairwise sums (§4)	Used; consistent with A30 doubled-additive structure.
ZS-A30 budget (6,38,83)/121	independent rank target (§11)	Inherited; used as target outside the measure (anti-circularity).
ZS-M4 $\alpha_{\text{BK}} = -\ln z^* $	clarified vs Koenigs λ^* (§6)	No conflict: α_{BK} and $\lambda^* = f'(z^*)$ are different objects.
ZS-F30 $\rho_K(\theta)$, g_K , seam $\rho_K(\theta+\pi) = -\rho_K(\theta)$	Seam Transport Theorem (§5)	Uses F30’s own objects; corrects/strengthens the holonomy claim; F14-route retracted.
ZS-F14 (ϵ, θ) joint ODE	identified as NOT the support-monodromy route	Conflict in v1.2 removed: F14 is radial–angular, not a linear support ODE.
ZS-S14 master action	S_{carrier} (tetrad form, §2.5)	Strict extension; reduction keeps B, W, G.
ZS-A17 metric No-Go	metric imported (§2)	Respected — no metric emergence claimed.
ZS-A19/A20 Brown dust; ZS-A28 top-form	matter frame, clock, vacuum (§2.5)	Used at DERIVED; field equations restored.

§16. Falsification gates

Gate	Layer	Trigger
F-F31.1	Mathematical (verified)	If the jump-only generator failed $\Phi_0 = \text{id}$, complete positivity, trace preservation, the intertwining $E_D \mathcal{L}^\star = \mathcal{L}_\pi E_D$, or uniqueness of the tracial stationary state, Theorem P1 fails. (Machine-verified; Appendix A.)
F-F31.2	Mathematical (verified)	If $\text{Spec}(\mathcal{L}^\star) \neq \text{Spec}(\mathcal{L}_\pi) \cup \{-\frac{1}{2}(\Gamma_c + \Gamma_d)\}$ (e.g. some coherence is not an eigenvector), Theorem P2 fails. (Verified to dimension 121; Appendix B.)
F-F31.3	Mathematical (verified)	If $U_K(\theta+\pi, \theta) \neq g_{K^2}$ or its singular values $\neq e^{\pm \rho_K}$, the Seam Transport Theorem R2a fails. (Verified; Appendix C.)
F-F31.4	Theoretical (OPEN target)	If the physical connection A_Z is not gauge-equivalent to A_K , R2b fails and the transport coordinate $\rho_{\text{tr}} = \rho_K$ does not promote to the physical closed-loop holonomy.
F-F31.5	Theoretical (OPEN target)	If no modular eigen-partial-isometries $v_{\{ab\}}$ exist in the Type II _i core (P1-core), the continuous-core embedding fails; if the modular spectrum gives $-\ln \lambda^* /\omega_Z \neq \arg \lambda^*/v_Z$, C_{int} fails.
F-F31.6	Theoretical	If the causal-Z measure were not normalizable — i.e. if $D(q_0 q^*)$ diverged or N_{pix} were unbounded — §10.2 fails. (The entropy budget makes this sharp.)

F-F31.7	Observational (terminal)	If, after fixing the clock map and B3, $T^* = \text{argmax } W_Z$ gave $\Omega_A(T^*)$ outside the pre-registered $ G < 10^{-3}$, the causal-entropic resolution is rejected; or if precision cosmology drove $\Omega_A,0$ far from 0.686 or established $w \neq -1$, the inherited budget is rejected.
F-F31.8	Anti-circularity	If 83/121 (or the rank budget) were found to have entered W_Z , the coincidence test is void.
F-F31.9	Anti-overclaim	If this paper is read as solving the coincidence problem (a DERIVED-CONDITIONAL gate), as closing R2b/C_int/Gate E (OPEN), as the full coherence-resolved lift (only the population sector is closed), or as deriving the metric (A17) or the absolute scale (B3), it is falsified by its own scope.

§17. Conclusion

v1.3 turns one repaired theorem into three independent contributions and removes an unforced parameter, while restoring the integrative master-action content. First, the jump-only modular GKLS generator yields the exact 121-face Lindblad spectrum $\text{Spec}(\mathcal{L}\star) = \text{Spec}(\mathcal{L}_\pi) \uplus \{-\frac{1}{2}(\Gamma_c + \Gamma_d)\}$ in closed form (Theorem P2), verified against the dense superoperator to dimension 121 — a genuine spectral theorem, not a toy check, and now strictly parameter-free since the dephasing term is gone. Second, the retracted F14-monodromy route is replaced by the exact seam half-transport $U_K = g_K^2 = \exp[\rho_K n \cdot \sigma]$ (Theorem R2a), whose singular-value Cartan length is $\ell_K = |\rho_K|$ and whose signed oriented rapidity $\rho_{\text{tr}}^{\wedge\{\text{oriented}\}} = \rho_K$ is recovered from the fixed axis; this is an open-path transport realization, not a closed-loop holonomy, so the promotion to a physical A_Z holonomy (R2b) remains open. Third, the causal-information present measure is provably normalizable by an entropy budget, $\int N_{\text{pix}} \sigma_Z \leq N_{\text{max}} D(\varrho_0 \| \varrho^*) < \infty$, with no MLSI — and the bulk–boundary relation is honestly only a HYPOTHESIS-strong capacity bound (data processing gives a value inequality, not a pointwise rate inequality), the coincidence test an explicitly conditional gate.

The honest terminus is a small, named set of computations and one measurement: R2b (gauge equivalence), P1-core (continuous-core isometries), C_int (spectral matching), the clock map $t = t(T)$, Gate E saturation, and the coincidence test G — together with the standing OPENs (the full coherence-resolved lift, the absolute scale B3, the imported metric A17). Each is concrete. Relative to every previous version, v1.3 is both more rigorous (three proved theorems, one retraction, picture-clean, parameter-free generator) and more complete (the carrier action and field equations restored). Zero new fitted parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED.

Acknowledgements & Code Availability

This paper consolidates internal Z-Spin Collaboration deep-exploration notes and three rounds of adversarial review of ZS-F31. The companion script `zs_f31_verify_v1_4.py` reproduces the load-bearing checks: the corrected z^* digits and Planck deviations ($\Omega_A 0.17\sigma$, all three $< 0.5\sigma$); the jump-only genuine GKLS lift in both pictures, with $\Phi_0 = \text{id}$, complete positivity, trace preservation/unitality, modular covariance, the intertwining $E_D \mathcal{L}\star = \mathcal{L}_\pi E_D$, and a unique tracial stationary state; a regression check that the retracted v1.1 family has $\Psi_0 = E_D \neq \text{id}$; the full 121-face spectrum $\text{Spec}(\mathcal{L}\star) = \text{Spec}(\mathcal{L}_\pi) \uplus \{-\frac{1}{2}(\Gamma_c + \Gamma_d)\}$ verified against the dense superoperator up to dimension 121 and exhibited in closed form at 14641; the sign-corrected Seam Transport (unsigned length $\ell_K = |\rho_K|$ and signed recovery $\rho_K = \frac{1}{2}\text{Tr}[(n \cdot \sigma) \log U_K]$, at BOTH signs); the Koenigs multiplier (with the honest note $-\ln|\lambda^*| \neq \alpha_{BK}$); the entropy-budget normalizability $\int_0^\infty \sigma_Z = D(\varrho_0 \| \varrho^*)$; and the Gate-E rationale check that value ordering does not imply rate ordering. The script prints an explicit OPEN registry (R2b,

P1-core, C_int, clock map, Z-screen, Gate E [HYPOTHESIS-strong], coincidence gate) that it does not count as PASS, so its summary reads ‘42/42 arithmetic, symbolic, and construction checks PASS; the operator gates above are not machine-verified.’ It contains no fail-open clause. This work used AI tools for verification and drafting; the author assumes full responsibility for all content.

Appendix A. The jump-only GKLS generator in both pictures: the five-property proof

Schrödinger $\mathcal{L}\star(\rho) = \sum_{a \neq b} r_{\{ab\}} (v_{\{ab\}} \rho v_{\{ab\}}^\dagger - \frac{1}{2} \{q_b, \rho\})$; Heisenberg $\mathcal{L}(x) = \mathcal{L}\star^\dagger(x) = \sum_{a \neq b} r_{\{ab\}} (v_{\{ab\}}^\dagger x v_{\{ab\}} - \frac{1}{2} \{q_b, x\})$; $v_{\{ab\}} = |a\rangle\langle b|$ on the faces, balanced symmetric rates $r_{\{ab\}} = r_{\{ba\}}$.

(1) $\Phi_0 = e^{\{0 \cdot \mathcal{L}\star\}} = \text{id}$. (2) $\mathcal{L}\star$ is in GKLS form (jumps $\sqrt{r_{\{ab\}}} v_{\{ab\}}$), so $\{\Phi_t\}$ is a normal CP semigroup. (3) $\text{Tr } \mathcal{L}\star(\rho) = \sum r_{\{ab\}} (\text{Tr}(v_{\{ab\}} \rho v_{\{ab\}}^\dagger) - \text{Tr}(q_b \rho)) = \sum r_{\{ab\}} (\text{Tr}(q_b \rho) - \text{Tr}(q_b \rho)) = 0$, equivalently $\mathcal{L}(I) = \sum r_{\{ab\}} (q_b - q_b) = 0$. (4) $\sigma_{s^\omega} v_{\{ab\}} = e^{\{is_\omega\{ab\}\}} v_{\{ab\}}$ (faces in the centralizer, $\omega_{\{ab\}} = \ln(h_a/h_b)$) gives $\sigma_{s^\omega} \mathcal{L}\star \sigma_{-s^\omega} = \mathcal{L}\star$ termwise, so $[\mathcal{L}\star, \sigma_{s^\omega}] = 0$. (5) For $c \neq d$, $v_{\{ab\}} |c\rangle\langle d| v_{\{ab\}}^\dagger = \delta_{\{bc\}} \delta_{\{bd\}} |a\rangle\langle a| = 0$, so the gain term annihilates coherences and $E_D \mathcal{L}\star (1 - E_D) = 0$; on the diagonal $\mathcal{L}\star$ restricts to $(\mathcal{L}_\pi p)_a = \sum_b (r_{\{ab\}} p_b - r_{\{ba\}} p_a)$, giving $E_D \mathcal{L}\star = \mathcal{L}_\pi E_D$. Uniqueness: for a strongly connected graph the only $\mathcal{L}\star$ -harmonic elements are scalars, so τ_{121} is the unique stationary state. ■

All of (1)–(5) and uniqueness are reproduced numerically on a 4-face register (Part B of the script), with the contrast that the v1.1 family $v \circ \Phi_t \circ E$ has $\Phi_0 = E_D \neq \text{id}$.

Appendix B. The full 121-face spectrum (Theorem P2): proof and verification

Coherence block. For $c \neq d$, the only surviving term of $\mathcal{L}\star(|c\rangle\langle d|)$ is $-\frac{1}{2} \sum_{a \neq b} r_{\{ab\}} \{q_b, |c\rangle\langle d|\} = -\frac{1}{2} (\sum_{a \neq c} r_{\{ac\}} + \sum_{a \neq d} r_{\{ad\}}) |c\rangle\langle d| = -\frac{1}{2} (\Gamma_c + \Gamma_d) |c\rangle\langle d|$, with $\Gamma_c = \sum_{a \neq c} r_{\{ac\}}$ the out-rate from c . Thus every coherence is an eigenvector with eigenvalue $-\frac{1}{2}(\Gamma_c + \Gamma_d)$. Population block. On the diagonal $\mathcal{L}\star = \mathcal{L}_\pi$, the classical generator with stationary uniform distribution and a spectral gap. Disjointness of the two invariant subspaces gives $\text{Spec}(\mathcal{L}\star) = \text{Spec}(\mathcal{L}_\pi) \cup \{-\frac{1}{2}(\Gamma_c + \Gamma_d) : c \neq d\}$. ■

Verification. For $N = 4, 8, 11$ the dense superoperator $\mathcal{L}\star$ (dimensions 16, 64, 121) is diagonalised and its spectrum matches the predicted union to 10^{-8} . For $N = 121$ the spectrum is exhibited from the formula: 121 population eigenvalues (under the doubled-additive A30 structure, the pairwise sums $\lambda_i + \lambda_j$ of the 11-face Laplacian, including one zero mode) and 14520 coherence eigenvalues $-\frac{1}{2}(\Gamma_c + \Gamma_d)$, totalling $14641 = 121^2$ — without constructing the 14641^2 matrix.

Appendix C. The Seam Transport Theorem (R2a): proof and verification

Let $g_K(\theta) = \exp[\frac{1}{2} \rho_K(\theta) n \cdot \sigma]$ with the fixed axis $n(\theta + \pi) = n(\theta)$ and the seam antisymmetry $\rho_K(\theta + \pi) = -\rho_K(\theta)$ (ZS-F30). Then $g_K(\theta + \pi) = \exp[-\frac{1}{2} \rho_K(\theta) n \cdot \sigma] = g_K(\theta)^{-1}$, so the open-path seam half-transport is $U_K(\theta + \pi, \theta) = g_K(\theta + \pi)^{-1} g_K(\theta) = g_K(\theta)^2 = \exp[\rho_K(\theta) n \cdot \sigma]$. Since $(n \cdot \sigma)^2 = I$ with eigenvalues ± 1 , U_K is Hermitian positive-definite with eigenvalues $e^{\{\pm \rho_K\}}$, hence singular values $\{e^{\{|\rho_K|\}}, e^{\{-|\rho_K|\}}\}$; the singular-value ratio gives the UNSIGNED Cartan length $\ell_K = \frac{1}{2}$

$\ln(s_{\max}/s_{\min}) = |\rho_K|$. The SIGNED, seam-odd rapidity is recovered from the principal logarithm $\log U_K = \rho_K n \cdot \sigma$ via $\rho_K = \frac{1}{2} \text{Tr}[(n \cdot \sigma) \log U_K]$ (using $\frac{1}{2} \text{Tr}[(n \cdot \sigma)^2] = 1$). Thus the oriented transport coordinate is $\rho_{\text{tr}}^{\{\text{oriented}\}} = \rho_K$. This is an open-path TRANSPORT statement; the singular values of an open transport are not gauge-invariant under independent endpoint gauge transformations, so the promotion to a physical closed-loop holonomy ρ_{hol} requires the gauge equivalence $A_Z \sim A_K$ (R2b). ■

Verification. For (ρ_K, n) at BOTH signs — $\{(+0.37, \text{generic}), (-0.37, \text{generic}), (+0.9, \hat{x}), (-0.15, \text{generic})\}$ — the script confirms $g_K(\theta + \pi) = g_K(\theta)^{-1}$, $U_K = g_K^2 = \exp[\rho_K n \cdot \sigma]$, the unsigned length $\frac{1}{2} \ln(s_{\max}/s_{\min}) = |\rho_K|$, and the signed recovery $\frac{1}{2} \text{Tr}[(n \cdot \sigma) \log U_K] = \rho_K$, all to machine precision (the negative cases are what catch the v1.3 sign slip). R2b ($A_Z \sim A_K$) is left OPEN.

Appendix D. Koenigs multiplier and the retracted F14 route

Koenigs (C_{int}). For $f(z) = i^z = e^{\{i\pi z/2\}}$, $f(z^*) = z^*$, the multiplier is $\lambda^* = f'(z^*) = (i\pi/2) z^*$, with $|\lambda^*| = (\pi/2)|z^*| = 0.89151 < 1$ (attracting), $-\ln|\lambda^*| = 0.1148346$, and $\arg \lambda^* = \pi/2 + \arg z^* = 2.2592496$. Koenigs' theorem gives an analytic χ near z^* with $\chi \circ f = \lambda^* \cdot \chi$. Matching one iteration to one modular step needs $-\ln|\lambda^*|/\omega_Z = \arg \lambda^*/v_Z$ (C_{int} , OPEN). The per-step contraction $-\ln|\lambda^*| = 0.1148346$ is distinct from $\alpha_{\text{BK}} = -\ln|z^*| = 0.5664173$.

Retracted F14 route. ZS-F14's joint ODE governs the (ε, θ) radial–angular pair (radial Lyapunov decay; angular Goldstone accumulation; conservation $Q = a^3 \varepsilon^2 \theta = A$; centrifugal boundary), not a linear support ODE with two independent periodic solutions h_1, h_2 . A fundamental matrix built from such h_i would have 2π -periodic columns and trivial loop monodromy. The v1.2 ‘F14-ODE monodromy target’ is therefore withdrawn in favour of the seam half-transport (Appendix C).

Appendix E. Verification ledger (v1.3)

Mirrors `zs_f31_verify_v1_4.py`. PASS/FAIL are arithmetic/symbolic/construction; the OPEN registry lists operator gates not machine-verified.

Check	Statement	Result
A.4 z^*	$ z^* = 0.5675551633$; $-\ln z^* = 0.5664173303 = y^* \pi/2$; $ z^* ^2 = 0.3221188634$	PASS
A.7 Planck	$\Omega_b 0.48\sigma$, $\Omega_{\text{cdm}} 0.01\sigma$, $\Omega_\Lambda 0.17\sigma$; $\Omega_\Lambda < 0.2\sigma$ AND all three $< 0.5\sigma$	PASS
B.1 $\Phi_0 = \text{id}$	jump-only $\Phi_0 = e^{\{0\mathcal{L}\star\}} = \text{identity}$	PASS
B.2 CP	$\text{Choi}(\Phi_{\{0.6\}})$ PSD	PASS
B.3 TP/unital	$\text{Tr } \mathcal{L}\star(\rho) = 0$ and $\mathcal{L}(I) = 0$ (both pictures)	PASS
B.4 intertwine	$E_D \mathcal{L}\star = \mathcal{L}_\pi E_D$ and $E_D \mathcal{L}\star(1 - E_D) = 0$	PASS
B.5 stationary	unique tracial stationary I/N ; $\dim \ker = 1$	PASS
B.6 covariance	$[\mathcal{L}\star, \sigma_s^\omega] = 0$ (faces in centralizer)	PASS
C.1 regression	v1.1 $\Psi_0 = E_D \neq \text{id}$; v1.3 $\Phi_0 = \text{id}$	PASS
D Koenigs	$ \lambda^* = 0.89151 < 1$; $-\ln \lambda^* = 0.1148346 \neq \alpha_{\text{BK}} = 0.5664173$; $\arg \lambda^* = 2.2592496$	PASS
E entropy budget	$\int_0^\infty \sigma_Z = D(q_0 \ q^*) - D(q_\infty \ q^*) = D(q_0)$; bounded (no MLSI); gap tail $1/(2\Delta)$	PASS
F S14 reduction	zero new structure, keep B,W,G $\rightarrow$ S14	PASS

G Seam Transport	$U_K = g_K^2 = \exp[\rho_K n \cdot \sigma]; \ell_K = \frac{1}{2} \ln(s_{\max}/s_{\min}) = \rho_K $ (both signs); signed $\rho_K = \frac{1}{2} \text{Tr}[(n \cdot \sigma) \log U_K]$ (R2a)	PASS
I Gate-E rationale	counterexample $e^{\{-T\}} \geq e^{\{-2T\}}$ with reversed rates at $T=0$: value ordering $\nRightarrow$ rate ordering $\Rightarrow$ Gate E is HYPOTHESIS-strong	PASS
H full spectrum	$\text{Spec}(\mathcal{L}\star) = \text{Spec}(\mathcal{L}_\pi) \cup \{-\frac{1}{2}(\Gamma_c + \Gamma_d)\}$; $N=4,8,11$ dense match; 121 = 14641 closed form (P2)	PASS
OPEN registry	R2b, P1-core, C_int, clock map $t(T)$, Z-screen, Gate E (inequality), Gcoin (B3-dependent)	OPEN

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Version History

v1.4 (March 2026) — Final internal version. Four final patches following a third adversarial review, all expressing the existing results more precisely (no new research). (1) Seam transport SIGN: the singular-value ratio gives the UNSIGNED Cartan length $\ell_K = |\rho_K|$; the signed, seam-odd rapidity $\rho_K = \frac{1}{2}\text{Tr}[(n \cdot \sigma)\log U_K]$ is recovered from the oriented axis, now tested at BOTH signs (correcting a v1.3 slip that read the ratio as ρ_K and tested only positive cases). (2) Transport vs holonomy: R2a is a canonical seam-TRANSPORT realization theorem (open path), not a closed-loop physical-holonomy theorem; the connection is $\text{sl}(2, \mathbb{C})$ -valued pure-gauge (a boost, fixed axis), and the promotion $\rho_{\text{tr}} \rightarrow \rho_{\text{hol}}$ is R2b (OPEN). (3) Gate E is downgraded from DERIVED-CONDITIONAL to HYPOTHESIS-strong, since data processing gives a value inequality $D_Z \geq D_X$, which does not imply a pointwise production-rate inequality (value ordering $\neq$ derivative ordering). (4) §2.5 carrier sector corrected: S_{BK} is Brown–Kuchař DUST (clock/frame), the nonminimal prefactor F belongs to S14, and S_{Λ} is a 3-form potential A_3 with $F_4 = dA_3$; the three displayed equations are the carrier-sector (not the full) Euler–Lagrange system. A results-architecture table is added; the spectrum is stated as a multiset union and framed as a general N -face abstraction of ZS-A30’s graph-specific computation. The three PROVEN contributions (P1, P2, R2a) and the inherited budget are unchanged. No new fitted parameter; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED. The OPEN items (R2b, P1-core, C_{int} , B3, Gate E) are deferred to ZS-F32 / ZS-A31.

v1.3 (March 2026): Three independent contributions — Theorem P2 (exact 121-face Lindblad spectrum), Theorem R2a (exact seam transport, replacing the RETRACTED F14-monodromy route), and an entropy-budget normalizability proof; the A/Q dephasing term removed (jump-only, parameter-free); both pictures stated; ‘central’ corrected to ‘modular centralizer’; the carrier action and field equations restored. [Superseded by v1.4: the seam result is sign-corrected and reclassified as transport (not holonomy); Gate E is HYPOTHESIS-strong; §2.5 carrier labels corrected.]

v1.2 (March 2026): Replaced v1.1’s broken population lift ($\Psi_0 = E_D \neq \text{id}$) by a genuine GKLS generator (Theorem P1); reduced the geometric gates; replaced the circular rank-relative-entropy present selection by a causal-diamond Z-Spin entropy measure. [Superseded in part by v1.3: the F14-monodromy route is retracted, the dephasing term removed, and P1 raised to the full spectrum P2.]

v1.1 / v1.0 (March 2026): Initial assembly of ZS-F30 and ZS-A30 into Γ_{CR} ; the Diagonal-in-Centralizer Lemma (PROVEN, retained); proper-time exclusion (retained, strengthened in §10). [v1.1’s $\iota \circ \Phi_{\text{t}} \circ E$ lift and its over-closed Theorems R/C/P are retracted.]